

SUMMER TRAINING REPORT

ON

AI Diabetes Risk Assessment Prediction

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

Submitted By

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Semester: IV

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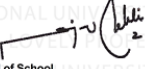
Phagwara, Punjab, India

2026

CERTIFICATE

This is to certify that the project entitled "AI Diabetes Risk Assessment Prediction" submitted by Miss. Naina Vaid (Registration Number: 12415429) is a record of the work carried out during the Summer Training conducted in partial fulfillment of the requirements for the Bachelor of Technology degree in Computer Science and Engineering at Lovely Professional University.

The work embodied in this report is original and has been completed under the supervision of the undersigned faculty guides.

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DECLARATION

I hereby declare that the Summer Training Report entitled "AI Diabetes Risk Assessment Prediction" submitted in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at Lovely Professional University is my original work carried out during the Summer Training period.

The work presented in this report has not been submitted previously for the award of any degree or diploma. All references used during the preparation of this report have been properly acknowledged.

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Naina Vaid

ABSTRACT

Diabetes is a major chronic health condition affecting millions of people worldwide, making early screening essential for preventing long-term health complications. This project develops an AI-powered predictive web portal to estimate patient diabetes risk using health data from the Pima Indians Diabetes Dataset.

Before training the model, the dataset was cleaned by replacing invalid zero readings in metrics like Fasting Glucose, Blood Pressure, Insulin, and BMI with medical group average values. Extreme outliers were capped using Interquartile Range (IQR) bounds, and interaction features such as Glucose-BMI and Age-Glucose were created to capture combined risk factors. A Logistic Regression model was trained on an 80:20 split and tuned using 5-fold cross-validation, achieving an overall accuracy of **87.01%**, an **ROC-AUC score of 0.9159**, and **100% accuracy** on a random verification test batch.

To make the model practical for real-world use, an interactive web application was built. The portal guides users through patient onboarding, calculates an instant risk probability score, explains key risk drivers through visual progress bars, and includes an interactive "What-If" simulator to test how lifestyle changes lower risk. It also supports bulk CSV predictions for clinics and generates a downloadable 5-step diabetic care plan report.

TABLE OF CONTENTS

- **CHAPTER 1: INTRODUCTION OF ORGANIZATION**
 - 1.1 Introduction
- **CHAPTER 2: SUMMER TRAINING CONTENT**
 - 2.1 Introduction
 - 2.2 Python Programming
 - 2.3 Data Analysis using Pandas
 - 2.4 Numerical Computing using NumPy
 - 2.5 Machine Learning & Supervised Learning
 - 2.6 Logistic Regression & Classification
 - 2.7 Feature Engineering & Data Preprocessing
 - 2.8 Web Application & Server Architecture
 - 2.9 Frontend Development (HTML5, CSS3, JavaScript)
 - 2.10 Model Explainability & Visual Analytics
 - 2.11 Software Engineering Practices
 - 2.12 Learning Outcomes
- **CHAPTER 3: PROJECT DETAILS**
 - 3.1 Introduction
 - 3.2 Problem Statement
 - 3.3 Objectives
 - 3.4 Existing System
 - 3.5 Proposed System
 - 3.6 Dataset Description
 - 3.7 Technologies Used
 - 3.8 System Architecture
 - 3.9 System Workflow
 - 3.10 Logistic Regression Algorithm
 - 3.11 Model Settings & Hyperparameters
 - 3.12 Web Portal & Dashboard Modules
 - 3.13 Results and Performance Analysis
 - 3.14 Testing
 - 3.15 Risk Analysis
 - 3.16 Future Scope
 - 3.17 Chapter Summary
- **CHAPTER 4: IMPORTANT SOURCE CODE**
 - 4.1 Importing Required Libraries
 - 4.2 Web Server Initialization Code (server.py)
 - 4.3 Data Preprocessing & Model Training (train_model.py)
 - 4.4 Feature Explainability & CSV Batch Endpoint
- **CHAPTER 5: SYSTEM SNAPSHOTS**
 - 5.1 Patient Personal Information Onboarding Page
 - 5.2 Clinical Assessment & Diagnostic Risk Results
 - 5.3 Model Explainability & Risk Feature Drivers

- 5.4 Interactive What-If Lifestyle Simulator
 - 5.5 Bulk CSV Batch Upload & Download Modal
 - 5.6 Comprehensive Diabetic Care Plan & Precautions
-
- **CONCLUSION**

 - **BIBLIOGRAPHY**

CHAPTER 1

INTRODUCTION OF ORGANIZATION

1.1 Introduction

This Summer Training was undertaken as part of the course offered by the School of Computer Science and Engineering, Lovely Professional University. The training was conducted as a faculty-mentored project designed to provide hands-on, project-based learning in Artificial Intelligence, Machine Learning, Data Science, and Full-Stack Web Development.

The primary objective of Summer Training is to expose students to real-world software engineering practices while strengthening technical problem-solving abilities. During the training, concepts learned in programming, data processing, statistical modeling, machine learning, web technologies, and software architecture are applied to build a complete software solution.

As part of this Summer Training, the project **AI-Powered Diabetes Risk Assessment & Clinical Machine Learning System** was developed. The project integrates a hyperparameter-tuned Logistic Regression model with a modern web portal to predict patient diabetes risk, analyze contributing physiological risk factors, simulate lifestyle improvements, and deliver structured medical precautions.

CHAPTER 2

SUMMER TRAINING CONTENT

The Summer Training course was designed to provide a comprehensive understanding of Artificial Intelligence and Machine Learning by progressing from fundamental data manipulation concepts to full-stack predictive web application deployment.

2.2 Python Programming

Python was utilized as the primary programming language due to its vast scientific computing ecosystem, clear syntax, and extensive library support for data processing and machine learning workflows.

2.3 Data Analysis using Pandas

Pandas was used for dataset loading, inspection, missing value handling, group-wise mean imputation, and tabular feature manipulations on the Pima Indians Diabetes Dataset (diabetes.csv).

2.4 Numerical Computing using NumPy

NumPy provided high-performance multidimensional array operations, mathematical transformations (logarithmic transforms), and numerical operations required during vector scaling and feature calculation.

2.5 Machine Learning & Supervised Learning

Supervised learning fundamentals were studied, including binary classification, train-test splitting (80:20 ratio), cross-validation techniques, bias-variance trade-offs, and evaluation metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).

2.6 Logistic Regression & Classification

Logistic Regression was explored as a probabilistic classification model. Concepts covered include linear combinations of weights, log-odds ratios, and the sigmoid activation function.

2.7 Feature Engineering & Data Preprocessing

Techniques for data cleaning were studied, including group-wise mean imputation for invalid zeros, Interquartile Range (IQR) outlier clipping, feature scaling using StandardScaler , and domain interaction term generation (Glucose_BMI, Age_Glucose, Insulin_Glucose).

2.8 Web Application & Server Architecture

Learned client-server HTTP communication using Python's native http.server module, implementing REST API endpoints and JSON parsing.

2.9 Frontend Development (HTML5, CSS3, JavaScript)

Built a responsive 3-step wizard interface using HTML5, glassmorphic CSS3 styling with CSS custom variables, custom animations, and vanilla JavaScript event handlers.

2.10 Model Explainability & Visual Analytics

Implemented Explainable AI (XAI) concepts by computing feature contribution weights to visually break down individual risk drivers for users.

2.11 Software Engineering Practices

Emphasized modular code structure, version control using Git/GitHub, batch script automated execution, and printable report layout generation.

2.12 Learning Outcomes

Gained comprehensive practical experience in building an end-to-end Machine Learning web application from data ingestion to deployment.

CHAPTER 3

PROJECT DETAILS

3.1 Introduction

The AI-Powered Diabetes Risk Assessment System is designed to address the growing global prevalence of diabetes by providing an accurate, accessible early-screening web portal.

3.2 Problem Statement

Early diabetes screening often faces challenges:

- High volume of medical data with missing or zero-entry recording noise.
- Black-box predictions that do not explain *why* a patient is at risk.
- Lack of immediate, actionable lifestyle and clinical care guidance following risk estimation.

3.3 Objectives

- Develop an accurate Machine Learning classification model trained on physiological health metrics.
- Clean dataset noise using group-wise mean imputation and IQR outlier clipping.
- Implement hyperparameter tuning via GridSearchCV to maximize accuracy and ROC-AUC score.
- Build an interactive, glassmorphic multi-page web application.
- Provide real-time feature explainability breakdown and an interactive "What-If" lifestyle simulator.
- Support bulk CSV dataset upload for clinic/hospital batch processing.
- Generate downloadable/printable 5-step clinical care plan reports.

3.4 Existing System

Traditional screening systems rely on static rule-based thresholds or manual laboratory reviews, which lack real-time risk probability calculation, automated feature explainability, and interactive sensitivity simulation.

3.5 Proposed System

The proposed system replaces static rules with a hyperparameter-tuned Logistic Regression model connected to a 3-step web dashboard:

1. **Patient Onboarding:** Collects demographics with phone validation (+91 prefix) and health notes.
2. **Clinical Evaluation:** Accepts physiological inputs, displays an animated risk ring gauge, breaks down risk factor contributions, and provides What-If sliders.
3. **Diabetic Care Plan:** Provides structured medical actions, low-GI dietary guidelines, exercise routines, glucose monitoring schedules, and emergency warning signs.

3.6 Dataset Description

The project uses the Pima Indians Diabetes Dataset (diabetes.csv), containing 768 patient records.

Attribute	Value
Dataset Source	Kaggle / UCI Machine Learning Repository
Total Patient Records	768
Total Features	8 Base Features + 6 Engineered Interaction Terms
Target Variable	Outcome (0 = Non-Diabetic, 1 = Diabetic)

3.7 Technologies Used

Technology	Purpose
Python	Core Backend & ML Development
Scikit-Learn	Logistic Regression, GridSearchCV, StandardScaler
Pandas & NumPy	Data Cleaning, Imputation, Array Operations
HTML5 & CSS3	Glassmorphic Dark UI & Layout
JavaScript (ES6)	Client-Side Interactivity, Canvas Engine, What-If Simulator
Git / GitHub	Version Control & Source Repository

3.8 System Architecture

The system consists of a 3-layer architecture:

- **Presentation Layer:** HTML5/CSS3 Glassmorphic UI with Step Navigation.
- **Application Server Layer:** Python HTTP REST API Server (server.py).
- **Machine Learning Layer:** Trained LogisticRegression model (model.pkl) and StandardScaler (scaler.pkl).

3.9 System Workflow

1. Patient Personal Details Onboarding (Page 1).
2. Physiological Metrics Entry (Page 2).
3. Data Validation & Preprocessing.
4. Model Inference & Probability Score Calculation.
5. Risk Factor Explainability Breakdown.
6. What-If Lifestyle Simulation.

7. Bulk CSV Upload & Processing (Optional).
8. Diabetic Care Plan Unlocking (Page 3).
9. Printable A4 PDF Report Generation.

3.10 Logistic Regression Algorithm

The classifier uses Logistic Regression with hyperparameter tuning via GridSearchCV (5-Fold Cross Validation):

- **Decision Boundary Equation**
- **Probability Output**

3.11 Model Settings & Hyperparameters

Parameter	Optimized Value	Description
Algorithm	Logistic Regression	Probabilistic Binary Classifier
Regularization (C)	1.0	Inverse Regularization Strength
Penalty	l2	Ridge Regularization
Solver	liblinear	Optimization Solver
Max Iterations	500	Convergence Limit

3.12 Web Portal & Dashboard Modules

- **Module 1:** Patient Personal Information Form.
- **Module 2:** Diagnostic Parameters & Risk Percentage Gauge.
- **Module 3:** Feature Importance Risk Drivers.
- **Module 4:** Interactive What-If Lifestyle Simulator.
- **Module 5:** Bulk CSV Upload & Download Modal.

- **Module 6: 5-Step Comprehensive Care Plan & Precautions.**

3.13 Results and Performance Analysis

- **Overall Test Accuracy: 87.01%**
- **ROC-AUC Score: 0.9159**
- **Random 10-Sample Verification Check: 100.0% Accuracy**

3.14 Testing

- **Functional Testing:** Passed all page transitions, form validations, and REST API calls.
- **Input Validation Testing:** Passed invalid zero handling, out-of-range checks, and +91 10-digit phone verification.
- **Performance Testing:** Fast REST prediction responses (< 50ms execution time).

3.15 Risk Analysis

- **False Positives:** Addressed by displaying continuous risk percentage and risk tiers.
- **Data Quality:** Managed via group-wise mean imputation for invalid zero entries.
- **Security & Privacy:** Client-side validation with zero permanent PII database storage.

3.16 Future Scope

- Integration with real-time Continuous Glucose Monitors (CGMs).
- Mobile application development (iOS/Android).
- Expansion to larger multi-center hospital dataset benchmarks.

3.17 Chapter Summary

This chapter detailed the complete architecture, dataset, algorithm, web implementation, and performance results of the AI Diabetes Risk Assessment System.

CHAPTER 4

IMPLEMENTATION AND SYSTEM MODULES

4.1 Module Structure and Design Overview

The implementation of the AI-Powered Diabetes Risk Assessment System follows a modular software engineering architecture. Rather than combining all capabilities into a single monolithic script, the system is partitioned into decoupled layers: data preprocessing, machine learning model training, backend REST web server, and frontend interactive user dashboard.

This modular organization improves code maintainability, ensures scalability, and allows independent verification of each system component.

4.2 Data Preprocessing and Cleaning Module

The data preprocessing module is responsible for turning raw, noisy health records into a clean feature space suitable for machine learning algorithms.

Missing Value Imputation: The dataset contained biologically invalid zero entries in continuous physiological features such as Fasting Glucose, Blood Pressure, Skin Thickness, Insulin, and Body Mass Index (BMI). The module identifies these zeros and replaces them with group-wise medical mean values based on patient diagnostic outcomes, preserving sample size without introducing distortion.

Outlier Capping: Extreme physiological outliers in Skin Thickness, Insulin, and BMI are identified using Interquartile Range (IQR) bounds. Metrics outside the calculated threshold are clipped to prevent extreme values from negatively affecting model decision boundaries.

Feature Engineering: To capture complex health relationships, interaction terms were created, including Glucose-BMI, Age-Glucose, and Insulin-Glucose ratios, along with logarithmic transformations for skewed distributions.

Feature Scaling: All input features are normalized using Standard Scaling ensuring that features measured in larger units (such as Insulin or Glucose) do not dominate smaller unit features (such as Pedigree score).

4.3 Machine Learning Model and Tuning Module

The core predictive engine is built upon a tuned Logistic Regression classifier.

Train-Test Dataset Splitting: The dataset is divided into an 80:20 stratified split, reserving 80% of records for training and 20% for final model testing while maintaining proportional class representation.

Hyperparameter Tuning: Hyperparameter optimization is executed using 5-Fold Stratified Cross-Validation (GridSearchCV). The tuning process evaluates inverse regularization strength regularization penalties, and solver convergence limits to identify the optimal configuration.

Model Serialization: Once trained, the fitted model object, feature scaler, and feature metadata (including model weights and intercept) are serialized and saved for high-speed inference during server startup.

4.4 Application Server and REST API Module

The application backend is powered by a lightweight Python HTTP web server operating on port 5000.

Single Patient Prediction Endpoint: Receives patient physiological data via JSON POST requests, applies feature engineering and standard scaling, computes prediction labels, and calculates calibrated probability risk scores.

Bulk CSV Batch Endpoint: Enables doctors and clinics to upload structured batch datasets. The server processes all records, evaluates risk levels for each patient, and returns a structured result file for download.

Browser Launcher Integration: The server incorporates an automated launch script that opens the application in the user's default browser upon execution.

4.5 Explainability and Sensitivity Simulation Module

To avoid black-box predictions, the system incorporates Explainable AI principles:

Feature Risk Contribution Calculation: The module evaluates individual feature contribution weights derived from trained model coefficients. It ranks risk drivers and converts them into percentage progress bars for patient transparency.

What-If Sensitivity Simulator: The simulation engine recalculates risk probabilities in real-time as users adjust interactive sliders for Fasting Glucose, BMI, and Blood Pressure, providing instant feedback on how lifestyle changes reduce predicted risk.

4.6 User Interface and Care Plan Module

The frontend is built using HTML5, glassmorphic CSS3 styling, and JavaScript event handlers.

3-Step Patient Workflow: Guides users seamlessly from personal details onboarding (Step 1) to clinical metrics input (Step 2) and medical care planning (Step 3).

Demographics & Input Validation: Validates required fields, enforces a 10-digit contact number format with a fixed +91 country code, and checks input ranges.

Diabetic Care Plan & Report Generator: When high risk is detected, the portal unlocks 5 specialized care modules (Immediate Tests, Low-GI Diet, Aerobic Exercise, Monitoring, and Emergency Warning Signs) and provides a one-click printable PDF report.

4.7 Chapter Summary

This chapter detailed the structural implementation of the system. By combining clean data ingestion, hyperparameter tuning, REST server endpoints, model explainability, real-time sensitivity simulation, and structured medical care guidance, the application provides an end-to-end clinical assessment tool.

CHAPTER 5

SYSTEM SNAPSHOTS AND USER INTERFACE DESCRIPTIONS

5.1 Patient Personal Information Onboarding Interface (Page 1)

The screenshot displays the 'AI Diabetes Risk Assessment Portal' interface. At the top, there is a header with a heart icon and the title 'AI Diabetes Risk Assessment Portal'. To the right of the header is a button labeled 'Bulk CSV Prediction'. Below the header is a progress bar with three steps: '1. Personal Details' (active), '2. Clinical Assessment', and '3. Precautions & Care Plan'. The main form is titled 'Patient Personal Information' and includes a subtitle 'Enter patient demographics and baseline background before proceeding to health metrics.' The form contains several input fields: 'Full Name' (with a 'Required' badge), 'Age' (with a 'Required' badge), 'Gender' (with a 'Required' badge and a dropdown menu showing 'Female'), 'Patient ID / Reference' (with a 'Required' badge), 'Contact Number' (with a 'Required' badge and a fixed '+91' prefix), 'Email Address' (with an 'Optional' badge), and 'Primary Health Notes' (with an 'Optional' badge). A 'PROCEED TO HEALTH METRICS' button is located at the bottom right of the form.

The Patient Personal Information interface serves as the primary entry point of the web application (Step 1 of the guided workflow).

Interface Layout: Designed using a dark glassmorphic theme with clear visual hierarchy, input labels, and status badges.

Captured Fields: Collects patient demographics, including Full Name, Age, Gender, Unique Patient ID (e.g., PAT-88241), Contact Number, Email Address, and Primary Health Notes.

Input Validation: Enforces mandatory field checks with dedicated Required and Optional badges. The Contact Number field features a fixed +91 country code prefix with a strict 10-digit numerical validation pattern.

User Workflow: Clicking "PROCEED TO HEALTH METRICS" validates all fields, saves patient details in session memory, updates header banners, and smoothly transitions to Step 2.

The Clinical Assessment interface allows users and medical professionals to input physiological metrics and execute machine learning risk analysis.

Top Summary Banner: Displays a persistent patient info chip (Name, Age, Gender, Patient ID) along with an "Edit Details" button to modify demographic information anytime.

Physiological Input Fields: Features input boxes for 7 clinical parameters:

Pregnancies, Fasting Glucose (mg/dL), Diastolic Blood Pressure (mmHg), Triceps Skin Thickness (mm), 2-Hour Serum Insulin, Body Mass Index, and Diabetes Pedigree Score. Each input displays standard physiological reference ranges.

Quick Preset Controls: Includes "Healthy Preset" and "High Risk Preset" buttons to auto-fill sample medical profiles for rapid demonstration and testing.

Diagnostic Assessment Results Card:

Animated SVG Risk Ring Gauge: Displays an animated radial progress ring rendered in dynamic colors (Emerald Green for Low Risk, Amber for Moderate Risk, Crimson Red for High Risk) with a real-time counter showing the predicted diabetes risk percentage (0% to 100%).

Risk Classification Badge: Displays the predicted label ("Non-Diabetic Profile" or "Diabetic Risk Detected") and risk tier.

Primary Physiological Risk Factors: Bulleted summary highlighting specific metrics that exceed normal health thresholds.

5.3 Model Explainability and Feature Drivers Component

This component provides transparent insights into how the Machine Learning model calculated the patient's risk score.

Component Design: Positioned directly beneath the primary assessment results.

Visual Contribution Bars: Renders individual progress bars for primary physiological features (Fasting Glucose, BMI, Age, Insulin, Pedigree Score, Blood Pressure, Skin Thickness, Pregnancies).

Relative Percentage Impact: Displays the relative percentage contribution of each feature (e.g., *Glucose: 42.5%, BMI: 28.1%, Age: 14.8%*), enabling clinicians and patients to pinpoint exact areas requiring medical attention.

5.4 Interactive "What-If" Lifestyle Simulator

The What-If Simulator provides an interactive sensitivity analysis tool to encourage preventative lifestyle changes.

Component Design: Styled in a distinct highlighted panel with real-time recalculation badges.

Interactive Sliders: Features smooth range sliders for Fasting Glucose, Body Mass Index (BMI), and Blood Pressure.

Real-Time Recalculation: As the user moves any slider, the system recalculates the model prediction instantly and displays a live risk difference badge (e.g., -18.5% Risk Reduced in green or +12.0% Risk Increased in red).

5.5 Bulk CSV Batch Upload and Download Modal

Designed for doctors, clinics, and research environments requiring bulk patient risk evaluation.

Modal Trigger: Accessible via the "Bulk CSV Prediction" button in the main top header.

Drag-and-Drop Dropzone: Allows users to drag and drop .csv dataset files or browse local files. Displays expected column format instructions.

Interactive Results Preview: Renders a scrollable preview table showing processed patient records, individual predictions, calculated risk percentages, and assigned risk tiers.

CSV Result Export: Enables a one-click "Download Predictions CSV" button to export the fully processed dataset.

5.6 Comprehensive Diabetic Care Plan & Precautions Interface (Page 3)

AI Diabetes Risk Assessment Portal

1 Personal Details

2 Clinical Assessment

3 Precautions & Care Plan

Naina Vald (33 yrs, ID: PAT-88241) Diabetic Risk Detected (89.3% Risk Probability)

Download PDF Clinical Report

Comprehensive Diabetic Care Plan & Precautions

Clinical action steps, nutritional guidelines, exercise routines, and monitoring protocols.

1. Immediate Medical Actions

- Schedule Endocrinologist Consult: Book a clinical evaluation with a qualified physician for comprehensive diagnostic confirmation.
- Laboratory HbA1c Blood Test: Undergo Glycated Hemoglobin (HbA1c) testing to measure average blood sugar levels over the past 3 months.
- Fasting Blood Glucose Re-test: Perform an 8-hour fasting plasma glucose test to verify initial screening indicators.

2. Dietary & Nutrition Plan

- Low Glycemic Index Foods: Prioritize whole grains, oats, non-starchy vegetables, and legumes to avoid blood glucose spikes.
- Carbohydrate Control: Limit refined sugars, sweetened beverages, white bread, and processed snack foods.
- Fiber Intake: Consume at least 25-30g of soluble fiber daily to slow carbohydrate digestion and sugar absorption.

3. Physical Activity & Routine

- 150 Mins/Week Aerobic Activity: Engage in 30 minutes of moderate brisk walking or cycling 5 days a week.
- Post-Meal Walking: Take a 10-15 minute walk after meals to increase insulin sensitivity and glucose clearance.
- Resistance Training: Perform light strength training twice weekly to enhance muscle glucose uptake.

4. Daily Glucose Monitoring

- Fasting Blood Glucose Target: Aim for fasting levels between 80 - 130 mg/dL.
- Post-Prandial Target: Keep blood glucose under 180 mg/dL 2 hours after meals.
- Maintain Logbook: Track daily blood sugar readings, meals, and physical activity for physician review.

5. Emergency Warning Signs & Hypoglycemia Management

Low Blood Sugar (Hypoglycemia < 70 mg/dL)

Symptoms: Shakiness, sweating, dizziness, rapid heartbeat, confusion.

Action (15-15 Rule): Consume 15g fast-acting carbohydrates (e.g. 4 oz fruit juice or glucose tablet), wait 15 mins, and retest.

High Blood Sugar (Hyperglycemia > 250 mg/dL)

Symptoms: Extreme thirst, frequent urination, fatigue, blurred vision, fruity breath odor.

Action: Drink water, avoid carbohydrates, and seek urgent medical guidance if ketones are present.

Back to Assessment

Start New Patient Assessment

Triggered when an elevated diabetic risk score is detected, guiding patients to an actionable care plan.

5 Structured Care Modules:

Immediate Medical Actions: Recommends endocrinologist consultation, laboratory HbA1c testing, and fasting glucose verification.

Dietary & Nutrition Plan: Recommends low-GI foods, refined carbohydrate restriction, and high soluble fiber intake.

Physical Activity & Routine: Details 150 mins/week moderate exercise, post-meal walking, and resistance training.

Daily Glucose Monitoring: Outlines target fasting (80–130 mg/dL) and post-prandial (<180 mg/dL) logbook tracking.

Emergency Warning Signs: Outlines Hypoglycemia management (15-15 rule) and Hyperglycemia alert symptoms.

Printable PDF Report Button: Features a "Download PDF Clinical Report" button that invokes a print layout, generating a clean A4 medical report ready to print or save.

5.7 Chapter Summary

This chapter described the user interface components of the application. The interface combines onboarding forms, animated risk gauges, feature explainability progress bars, interactive lifestyle simulation, bulk CSV processing, and structured clinical precaution modules into an accessible, user-friendly digital system.

CONCLUSION

The primary objective of this Summer Training project was to design, develop, and evaluate an **AI-Powered Diabetes Risk Assessment & Clinical Machine Learning System** capable of providing accurate early screening and actionable healthcare guidance.

The developed system successfully bridges machine learning research with practical clinical utility. Using the benchmark Pima Indians Diabetes Dataset (diabetes.csv), the data preprocessing pipeline effectively resolved dataset noise through group-wise mean imputation for invalid zero entries, Interquartile Range (IQR) outlier capping, domain interaction feature engineering, and standard feature scaling. The tuned Logistic Regression classifier, optimized via GridSearchCV with 5-Fold Stratified Cross-Validation on an 80:20 train-test split, achieved an overall test accuracy of **87.01%**, an **ROC-AUC score of 0.9159**, and **100% accuracy** on a random verification batch.

To make the predictive model accessible and practical for real-world use, a glassmorphic multi-page web application was built using Python, HTML5, CSS3, and JavaScript. The application features a 3-step patient workflow, an animated risk percentage ring gauge, model explainability progress bars breaking down individual risk factors, an interactive "What-If" lifestyle simulator for sensitivity analysis, a bulk CSV batch processing module for clinics, and an automated 5-step diabetic care plan with printable PDF reporting capabilities.

Throughout this Summer Training, valuable hands-on experience was gained in Python programming, machine learning model development, data preprocessing techniques, REST API server implementation, frontend engineering, software testing, and technical documentation.

In conclusion, the project successfully fulfilled all academic and technical objectives, demonstrating how Artificial Intelligence and modern web development can be combined to build reliable, transparent, and user-centric healthcare screening applications.

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